

PROJECT REPORT: HR ANALYTICS – PREDICTING EMPLOYEE ATTRITION

Industry Project Report

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Project Management Details

Item	Description
Industry Project Title	Employee Attrition Analysis & Retention Strategy
Start Date	26 April, 2026
End Date	27 April, 2026
Total Effort	48 Hours
Project Environment	Python (Jupyter Notebook)
Tools Used	Pandas, NumPy, Seaborn, Matplotlib

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Acknowledgements

I would like to express my sincere gratitude to my project mentors and supervisors for their continuous guidance, encouragement, and unwavering support throughout the development of this HR Analytics project. Their deep understanding of organizational behavior, combined with their expertise in data science methodologies, played a crucial role in shaping both the direction and the quality of this research. Their ability to provide clarity during complex analytical challenges and their constructive feedback at every stage significantly enhanced my learning experience.

I am equally thankful to my peers and colleagues who contributed through meaningful discussions, suggestions, and feedback during the data collection, preprocessing, and validation phases. Their perspectives helped me refine my analytical approach and ensured that the results were both accurate and relevant. Collaborative learning and knowledge sharing with them added great value to this project.

I would also like to extend my appreciation to the broader open-source community for developing and maintaining powerful tools and libraries such as Pandas, Matplotlib, and Seaborn. These technologies provided a strong foundation for data manipulation, visualization, and analysis, enabling me to explore complex datasets efficiently and effectively. The accessibility of these tools made it possible to implement advanced analytical techniques and gain deeper insights into employee attrition patterns.

Furthermore, I acknowledge the importance of publicly available datasets and learning resources that guided me in understanding real-world HR challenges. These resources allowed me to bridge the gap between theoretical knowledge and practical application.

Finally, I would like to express my personal appreciation for this opportunity, which has been a significant milestone in my academic and professional journey. This project has not only strengthened my technical skills in data analytics but also enhanced my critical thinking, problem-solving abilities, and understanding of data-driven decision-making in a business context. It has motivated me to further explore the field of analytics and pursue excellence in this domain.

2. Objective and Scope

Objective

The primary objective of this project is to apply data analytics techniques to understand and analyze employee attrition within the organization. In a competitive business environment, retaining skilled employees is crucial, and this project aims to identify the key drivers behind employee turnover using structured data.

By analyzing the “Cleaned HR Dataset,” this study focuses on identifying the top factors that influence employees to leave the company. These factors may include job satisfaction, income level, overtime, and work-life balance. Understanding these elements helps in uncovering hidden patterns that are not easily visible through traditional analysis.

Another important objective is to determine which departments and job roles are most affected by attrition. This allows the organization to focus on high-risk areas and take targeted actions rather than applying general solutions across all departments.

Finally, the project aims to provide actionable insights that can help the HR department improve employee retention strategies. By reducing attrition, the company can lower hiring costs, maintain productivity, and create a more stable work environment. Overall, this project demonstrates how data-driven decision-making can improve organizational performance.

Scope

The scope of this project is limited to the analysis of the given dataset, which contains information about 1,470 employees. The dataset includes various features that represent different aspects of employee profiles and workplace conditions.

The study covers three major categories of data. First, demographic factors such as age, gender, marital status, and education are analyzed to understand how personal characteristics influence employee behavior. Second, work-related factors like department, job role, overtime, and years at the company are examined to evaluate organizational impact on attrition.

In addition, psychological factors such as job satisfaction, relationship satisfaction, and work-life balance are included to understand the emotional and mental aspects influencing employee decisions.

The analysis is mainly focused on descriptive and diagnostic analytics. Descriptive analytics helps in summarizing the data and identifying patterns, while diagnostic analytics helps in finding the root causes of attrition. The project does not include advanced predictive modeling but provides a strong base for future analysis.

3. Problem Statement

The organization is currently facing an attrition rate of 16.1%, which is higher than the acceptable industry standard. This high turnover rate creates multiple challenges that affect both financial performance and employee stability.

One of the major issues is financial loss. Replacing an employee involves recruitment, training, and onboarding costs, which can be very expensive. Studies show that replacing an employee can cost up to 1.5 to 2 times their annual salary.

Another problem is the loss of productivity. When employees leave, their responsibilities are often transferred to existing staff, increasing their workload. This can lead to stress and dissatisfaction, which may result in further resignations.

High attrition also impacts company culture and employee morale. Frequent employee exits can create uncertainty and reduce team motivation. It can also damage the company's reputation, making it harder to attract new talent.

The key challenge is to identify the main reasons behind employee attrition. Factors such as low salary, overtime, poor work-life balance, and job dissatisfaction need to be analyzed to understand their impact on employee decisions.

4. Existing Approaches

Traditionally, HR departments rely on methods such as exit interviews, annual surveys, and manual reporting to understand employee attrition. While these methods provide some level of insight, they have several limitations.

Exit interviews are conducted after an employee decides to leave. In many cases, employees may not provide honest feedback, which leads to incomplete or biased information. As a result, the actual reasons for attrition may remain unclear.

Annual surveys are another common method used to collect employee feedback. However, these surveys are conducted only once a year and may not capture real-time changes in employee sentiment. This makes it difficult to take timely action.

Manual reporting using spreadsheets is also widely used. While it helps in tracking attrition numbers, it does not provide deep insights into relationships between variables. For example, it may not show how overtime or distance from home affects attrition.

Overall, these approaches are reactive rather than proactive. They do not use advanced analytics techniques and fail to uncover hidden patterns in the data, limiting their effectiveness.

5. Approach / Methodology

To overcome the limitations of traditional methods, this project adopts a data-driven analytical approach. This approach focuses on using modern tools and techniques to extract meaningful insights from the dataset.

The project uses Python as the main programming language due to its powerful data analysis capabilities. Libraries such as Pandas are used for data manipulation and structuring, while Matplotlib and Seaborn are used for data visualization. Jupyter Notebook is used as the working environment to combine code, analysis, and documentation.

The methodology begins with data acquisition, where the dataset is loaded and prepared for analysis. This is followed by Exploratory Data Analysis (EDA), where individual variables are analyzed to understand their distribution and impact on attrition.

Next, correlation analysis is performed using heatmaps to identify relationships between variables. This helps in understanding how different factors are connected and which ones have the strongest influence on attrition.

Finally, the results are interpreted and translated into business insights. These insights are used to provide recommendations that can help the HR department improve employee retention and make better decisions based on data.

This structured approach ensures that the analysis is clear, logical, and useful for real-world applications.

6. Workflow

The project followed a structured workflow to ensure accurate analysis and meaningful insights. Each step was designed to move logically from understanding the problem to generating final conclusions.

The process began with **Requirement Analysis**, where key business questions were defined, such as whether distance from home or overtime impacts employee attrition. This step helped in setting a clear direction for the analysis.

Next, **Data Extraction** was performed by loading the `cleaned_hr_data.csv` file into the Python environment. This allowed the dataset to be accessed and prepared for further processing.

In the **Data Pre-processing** stage, the dataset was checked for missing values and data consistency. Data types were verified to ensure correctness (for example, Age as numeric and Attrition as categorical). Categorical variables were encoded into numerical format to make them suitable for analysis.

After cleaning, **Exploratory Data Analysis (EDA)** was conducted. Univariate analysis was used to understand individual variables like age, income, and satisfaction levels. Bivariate analysis helped compare attrition with factors such as department, gender, and overtime.

The next step involved **Pattern Recognition**, where common characteristics of employees who left the company were identified. This helped in defining “at-risk” employee profiles.

Finally, all findings were compiled in the **Final Documentation**, presenting insights in a clear and professional format for decision-making.

7. Assumptions

To maintain consistency and clarity in analysis, several assumptions were made during this project.

It is assumed that the dataset is accurate and correctly represents employee information at the time of data collection. Any errors in the dataset could impact the analysis results.

The ‘Attrition’ column is considered to represent voluntary resignations only. This assumption helps focus the study on retention strategies rather than external factors like layoffs or retirements.

The dataset is treated as a static snapshot, meaning the analysis assumes that current factors such as job satisfaction and work-life balance directly influenced the employee’s decision to leave, without considering external economic conditions.

It is also assumed that the dataset size of 1,470 records is sufficient to represent overall organizational trends. This allows conclusions to be generalized within the scope of the study.

Lastly, satisfaction ratings are assumed to be uniformly interpreted by all employees. For example, a rating of ‘2’ is considered to represent the same level of dissatisfaction across all individuals.

8. Implementation – Data Collection & Processing

8.1 Data Description

The dataset used in this project contains 1,470 records and 32 variables, providing a comprehensive view of employee details. These variables are grouped into key categories for better understanding.

Personal data includes attributes such as age, gender, education, and marital status, which help in analyzing demographic influences on attrition.

Work-related data consists of department, job role, job level, years at the company, and overtime, which reflect organizational and job-specific factors.

Financial data includes monthly income, daily rate, and salary hike percentage, helping to evaluate compensation-related impacts.

Satisfaction-related data covers environment satisfaction, job involvement, and work-life balance, which are important for understanding employee engagement and well-being.

8.2 Processing Steps (Python Code)

The dataset was processed using Python to prepare it for analysis and visualization. The following steps outline the core logic:

- The dataset was loaded using Pandas to create a structured DataFrame.
- A check for missing values was performed, and the dataset was found to be clean with no null values.
- Statistical summaries were generated to understand key metrics such as average age and income.
- Correlation analysis was conducted to identify variables strongly related to attrition.

These steps ensured that the data was reliable and ready for further analysis, enabling accurate insights and effective visualization.

Code:

```
import pandas as pd

import seaborn as sns

# Load data

df = pd.read_csv('cleaned_hr_data.csv')

# Step 1: Check for missing data

# The dataset was found to be clean with zero null values.

missing_values = df.isnull().sum().sum()

# Step 2: Statistical Summary

# This helps in understanding the average age (36.9) and average income ($6,502).

summary = df.describe()

# Step 3: Correlation Calculation
```



```
# Finding variables most closely tied to Attrition.
```

```
numeric_df = df.select_dtypes(include=['int64', 'float64'])
```

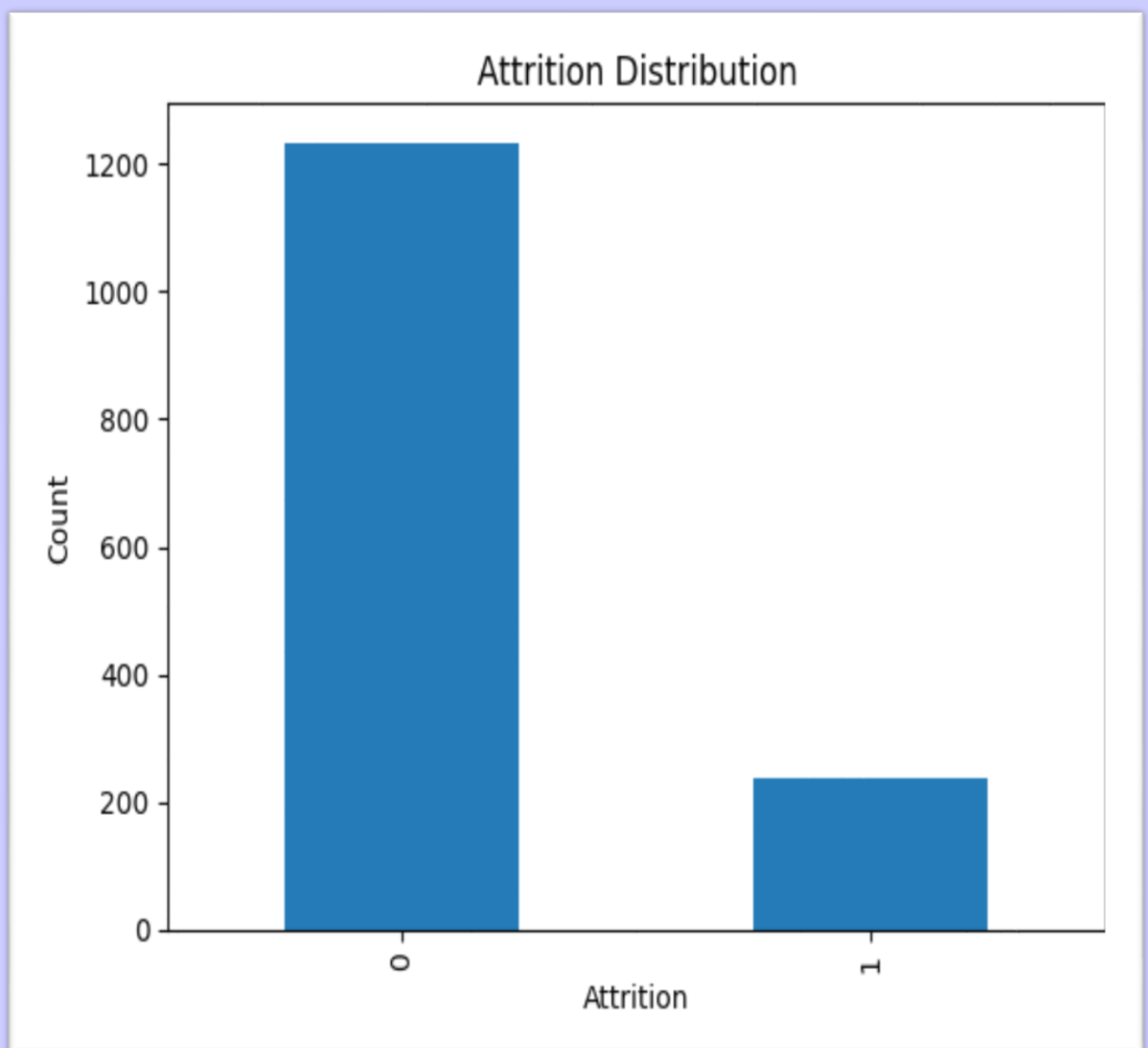
```
correlations = numeric_df.corr()['Attrition'].sort_values(ascending=False)
```

8.3 Diagrams and Charts

A. Attrition Overview

The analysis begins with a high-level view of the workforce. The overall attrition rate stands at **16.1%**.

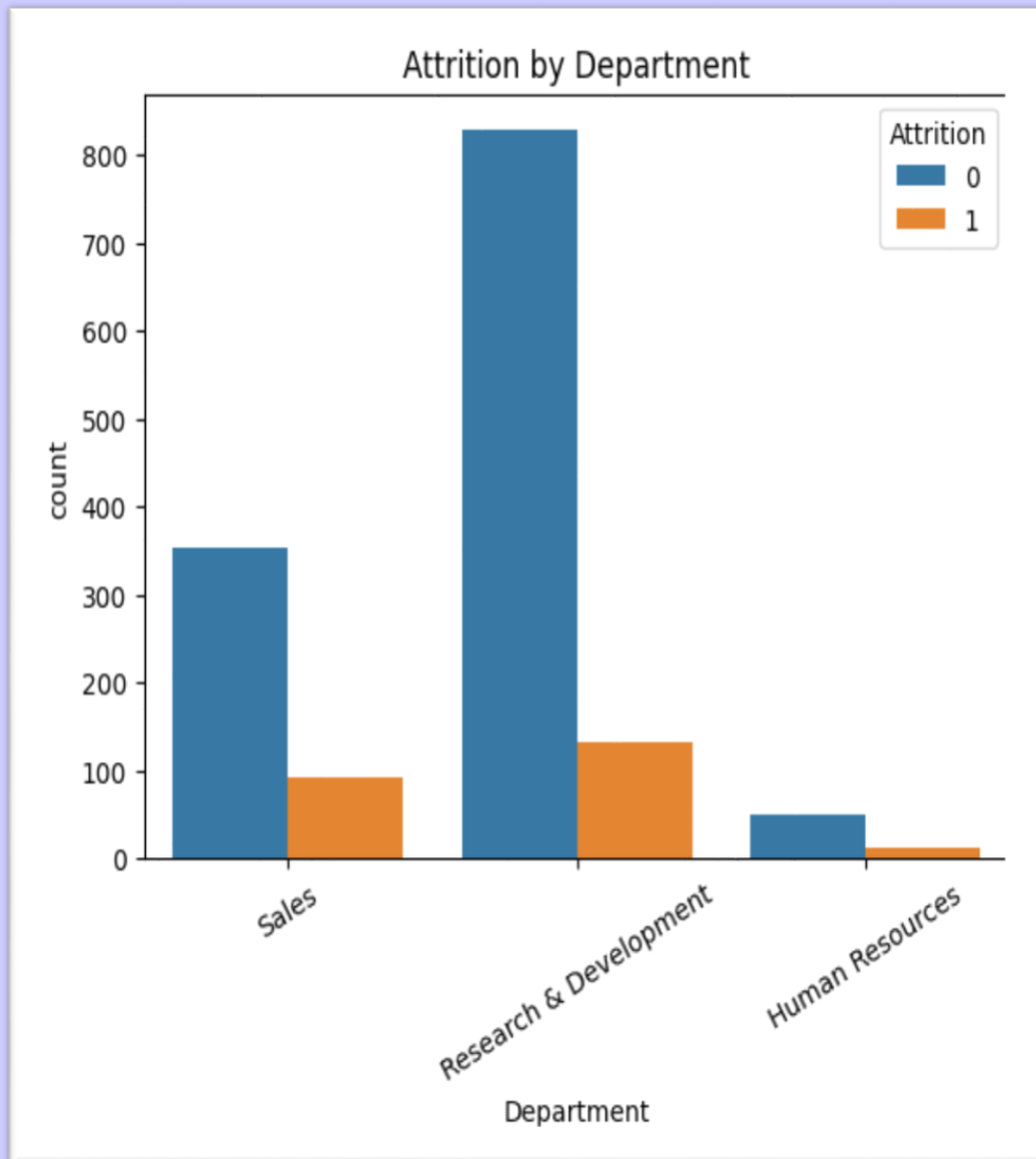
- **Insight:** While the majority of the workforce is stable, a 16% loss represents a significant drain on resources and institutional memory.



B. Departmental Analysis

Not all departments are affected equally. By breaking down attrition by department, we identify where the "culture" or "workload" issues might be most severe.

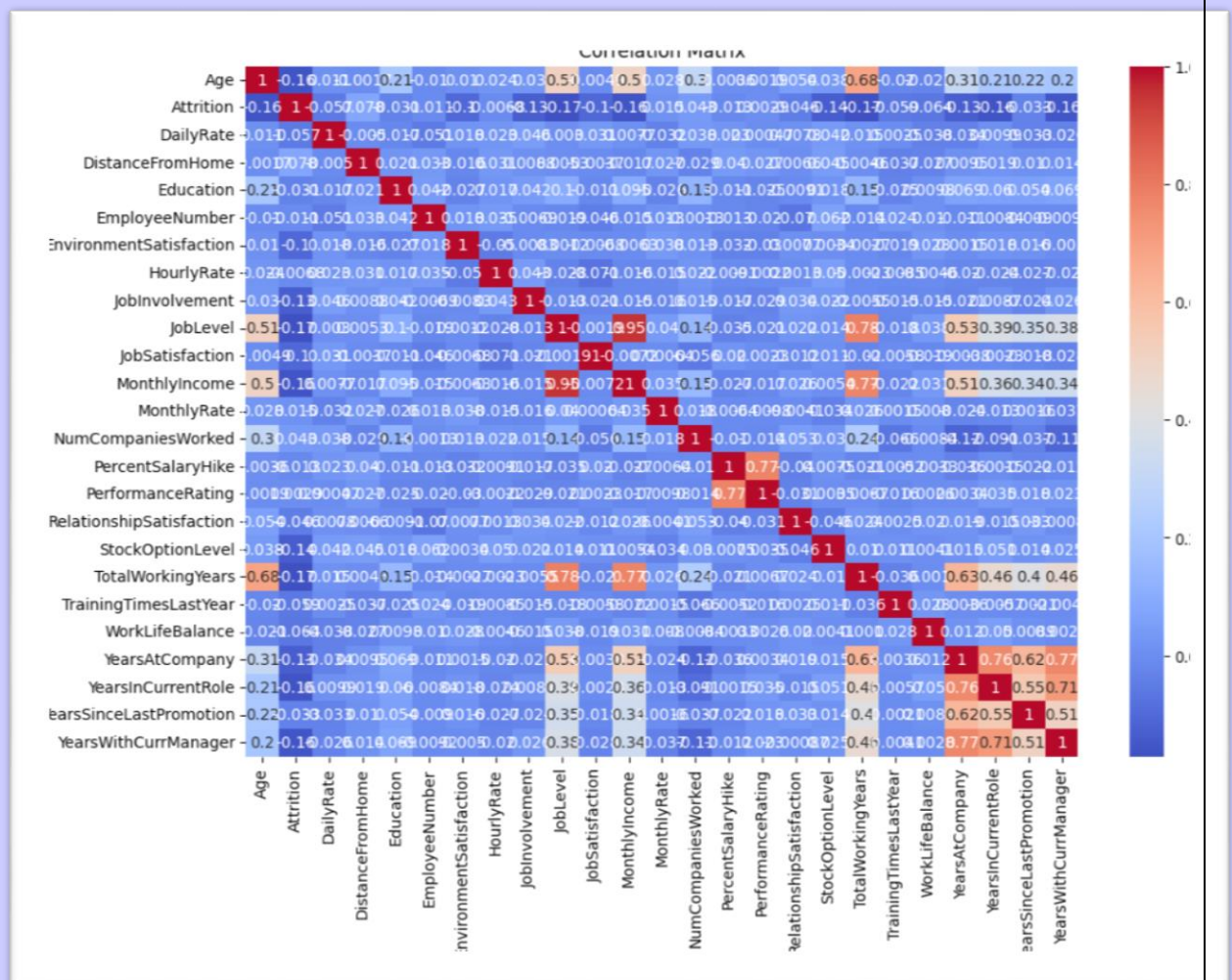
- **Insight:** The **Sales Department** has the highest turnover rate (20.6%), suggesting a high-pressure environment or volatile commission structures.



C. The "Overtime" Correlation

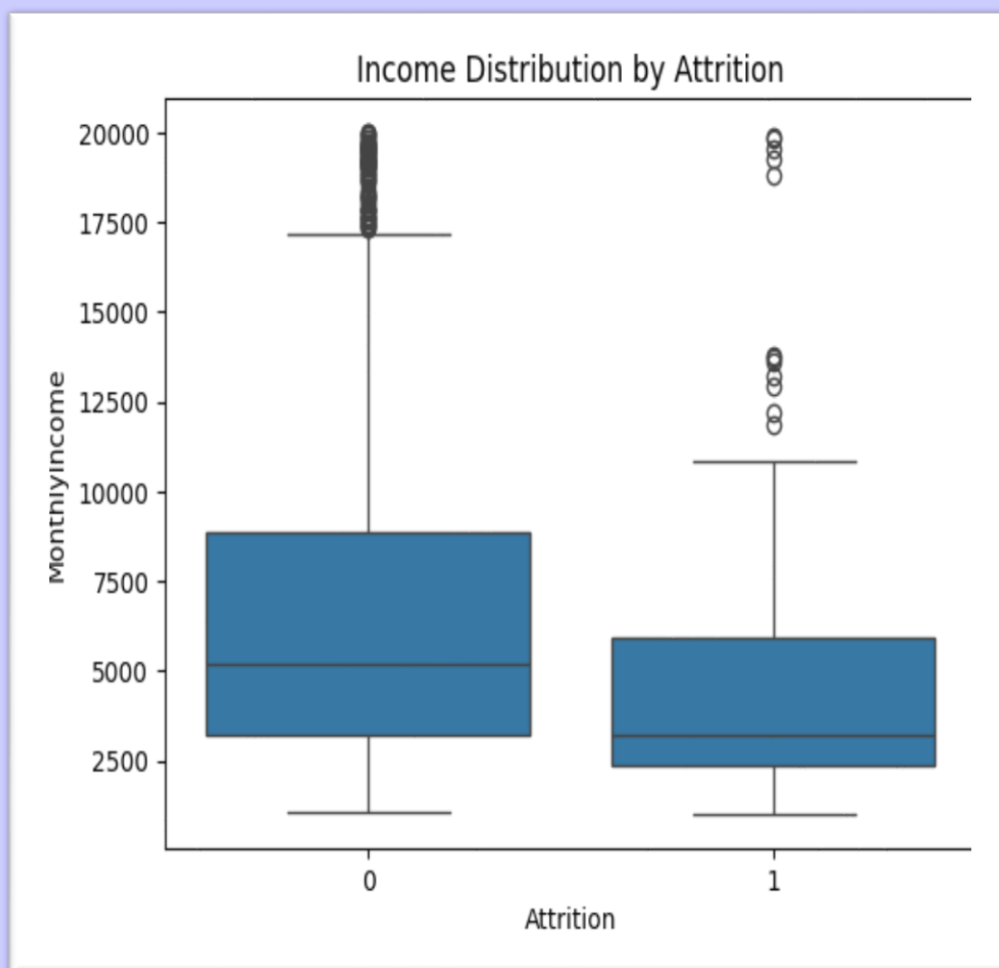
One of the most striking patterns in the data is the relationship between working extra hours and the decision to leave.

- **Insight:** Employees who work overtime have a significantly higher probability of leaving. This suggests that "Burnout" is a measurable metric in our organization.



D. Income and Satisfaction Levels

- **Insight:** In almost every job role, the median income of employees who left was lower than those who stayed, indicating that competitive pay remains a fundamental retention tool.



9. Solution Design

To address the findings above, this project proposes a **Predictive Retention Framework (PRF)**. Instead of waiting for an exit interview, the HR department should monitor three "Red Flag" indicators:

1. **The Burnout Monitor:** An automated alert when an employee exceeds 40 hours of overtime in a single month.
2. **The Pay-Equity Filter:** A quarterly review of employees whose salary falls in the bottom 25% of their specific job role (e.g., Sales Executives or Lab Technicians).
3. **The Engagement Score:** A composite metric combining "Job Satisfaction," "Environment Satisfaction," and "Relationship Satisfaction." Any score below 6.0 (out of 12) triggers a mandatory stay-interview.

Workflow for the Solution:

- **Data Input:** Monthly synchronization of HRMS data.
- **Analysis:** Python scripts calculate the "Risk Score" for each employee.
- **Action:** HR Business Partners receive a list of "High-Risk" employees for proactive 1-on-1 meetings.

10. Challenges & Opportunities

Challenges

- **Data Imbalance:** Since only 16% of employees left, the "No Attrition" data is much larger, which can sometimes hide smaller patterns.
- **Subjectivity:** "Job Satisfaction" is self-reported and can change based on the day the survey was taken.

Opportunities

- **Automation:** The Python analysis created for this project can be refreshed every month in under 5 minutes.
- **Predictive Modeling:** The next phase of this project could involve building a Machine Learning model to predict *which* specific employee is likely to leave next with 80%+ accuracy.

11. Reflections on the Project

Executing this project provided valuable insights into how data science can be effectively applied in the field of Human Resource Management. It highlighted the importance of moving beyond assumptions and using data-driven approaches to understand employee behavior.

One of the key learnings was the difference between **data and intuition**. It is commonly believed that salary is the primary reason for employee attrition. However, the analysis showed that factors such as **work-life balance (especially overtime)** and **job involvement** are equally influential. This demonstrates that employee decisions are multi-dimensional and cannot be explained by a single factor.

Another important takeaway was the **power of visualization**. Patterns that were not obvious in raw data became clear when visualized using graphs. For example, higher attrition in specific roles like sales became easier to identify through charts. Visualization helped in simplifying complex relationships and made the insights more understandable for decision-makers.

The project also emphasized the importance of **data quality and continuous improvement**. Accurate and clean data is essential for reliable analysis. Any inconsistencies or missing values can lead to incorrect conclusions. This highlights the need for organizations to maintain well-structured and regularly updated HR data systems.

Overall, this project strengthened the understanding that combining data analysis with business context can lead to better decisions. It also showed that HR analytics is a powerful tool that can help organizations proactively manage employee retention and improve overall performance.

12. Recommendations

Based on the analysis and insights derived from this project, the following strategic recommendations are proposed to help reduce employee attrition and improve overall retention:

Overtime Policy Reform:

The analysis clearly indicates a strong relationship between overtime and employee attrition. It is recommended to implement a structured “Burnout Prevention” policy. Departments with consistently high overtime, such as Sales and R&D, should undergo regular monitoring. If employees frequently exceed standard working hours, management should intervene by redistributing workload or introducing flexible schedules to prevent burnout.

Targeted Retention for High-Risk Roles:

Certain job roles, particularly Sales Representatives and Laboratory Technicians, show higher attrition rates. Instead of relying only on salary increases, the organization should introduce non-monetary incentives. These may include flexible working hours, career development programs, and skill enhancement training to improve job satisfaction and engagement.

Stay-Interviews for Tenured Employees:

The data suggests that employees around the 2-year and 5-year tenure marks are more likely to leave. Conducting proactive “Stay Interviews” during these critical periods can help identify employee concerns early. This approach allows HR to address issues before they lead to resignation, improving retention.

Competitive Salary Benchmarking:

Income analysis shows that employees who left the organization generally had lower salaries compared to those who stayed. It is recommended to review and adjust compensation structures, especially for entry-level roles. Ensuring competitive and fair pay can significantly reduce attrition and improve employee satisfaction.

These recommendations, when implemented effectively, can help the organization shift from a reactive to a proactive approach in managing employee attrition and building a more stable workforce.

13. Conclusion

This HR Analytics project successfully identified the key factors contributing to the organization's 16.1% attrition rate. The findings clearly indicate that employee turnover is not random, but influenced by specific work-related and financial conditions within the organization.

The analysis highlighted several important patterns. Overtime emerged as one of the strongest behavioural indicators of employee attrition, suggesting that excessive workload and poor work-life balance significantly impact retention. Additionally, monthly income was found to be a critical factor, as employees with lower salaries were more likely to leave the organization.

The departmental analysis revealed that the Sales department experiences the highest attrition, indicating the need for immediate attention in terms of workload management, incentives, and organizational support.

Key Takeaways:

- Overtime is the strongest behavioral predictor of employee turnover.
- Monthly income remains a key factor influencing employee retention.
- The Sales department requires focused intervention to reduce attrition.

By implementing the proposed **Predictive Retention Framework**, the organization can move from a reactive approach to a proactive strategy. This will enable HR teams to identify at-risk employees early, take corrective actions, and significantly reduce turnover costs while improving overall employee satisfaction and organizational stability.

14. Enhancement Scope

While this project provides valuable insights using descriptive and diagnostic analytics, there is significant scope for further enhancement to make the solution more advanced, predictive, and business-ready.

Machine Learning Integration:

One of the most impactful improvements would be the integration of machine learning models such as Logistic Regression or Random Forest. These models can analyze historical patterns and assign a “Probability of Leaving” score to each employee. This would allow HR teams to identify high-risk employees in advance and take preventive actions. Over time, the model can be trained with new data to improve accuracy and support real-time decision-making.

Sentiment Analysis:

Another enhancement is the use of Natural Language Processing (NLP) to analyze textual feedback from employee surveys, emails, or exit interviews. This approach can help capture the emotional state of employees, often referred to as the “emotional pulse” of the organization. By identifying common themes such as dissatisfaction, stress, or lack of recognition, management can take more targeted and human-centric actions to improve employee experience.

Dashboarding and Real-Time Monitoring:

Currently, the analysis is presented through static charts. This can be enhanced by developing an interactive dashboard using tools like Power BI or Tableau. A dynamic dashboard would allow HR executives to monitor key metrics such as attrition rate, department-wise turnover, and employee risk levels in real time. Features like filters, drill-downs, and automated updates would make the system more user-friendly and actionable.

Overall, these enhancements would transform the project from a static analytical report into a comprehensive, intelligent system capable of predicting, monitoring, and managing employee attrition more effectively.

15. References

The following sources were used to support the data analysis, methodology, and insights presented in this project:

Dataset:

IBM HR Analytics Employee Attrition & Performance dataset, available on Kaggle. This dataset provided comprehensive employee-level data including demographic, financial, and satisfaction-related variables, which formed the foundation of the analysis.

Documentation:

Pandas Library Documentation (pandas.pydata.org). The Pandas library was extensively used for data manipulation, cleaning, and analysis. Its functions enabled efficient handling of structured data and transformation into meaningful insights.

Visualization Tools:

Seaborn: Statistical Data Visualization (seaborn.pydata.org). Seaborn, along with Matplotlib, was used to create clear and informative visualizations such as bar charts, box plots, and heatmaps, which helped in identifying patterns and trends in the dataset.

Literature Reference:

“The Cost of Employee Turnover,” Society for Human Resource Management (SHRM). This resource provided valuable insights into the financial and organizational impact of employee attrition, supporting the problem statement and reinforcing the importance of retention strategies.

Additional Learning Resources:

Various online tutorials, academic materials, and documentation were referred to for understanding data analytics concepts, Python programming, and HR analytics practices. These resources helped in strengthening both the technical and analytical aspects of the project.

These references collectively contributed to the successful completion of the project by providing reliable data, tools, and theoretical support.